

1. Administrative & Study Identification

Full Study Title: A Phase IV Study of Safety and Efficacy of Everolimus in Taiwanese Patients With Tuberous Sclerosis Complex Who Have Renal Angiomyolipoma (TSC-AML) **Short Title/Acronym:** CRAD001M2402 TSC-AML PAC Study **Protocol Number:** CRAD001M2402 **NCT Number:** NCT05252585 **Posting ID:** 621134017657881556 **Sponsor / Company Name:** Novartis **Investigational Product:** Everolimus **Phase of Development:** Phase IV **Trial Status:** Active, Not Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety and efficacy of everolimus in Taiwanese patients with renal angiomyolipoma (AML) associated with tuberous sclerosis complex (TSC). **Secondary Objectives:** To evaluate the AML response rate, AML progression rate, and laboratory abnormalities.

2.2 Background & Rationale This open-label, prospective, single-arm, multicenter Phase IV post-approval commitment (PAC) study is conducted to generate safety and efficacy data for everolimus in the real-world clinical setting. Specifically, it aims to evaluate Taiwanese patients who have a confirmed diagnosis of TSC-AML and fulfill the local reimbursement criteria for targeted everolimus treatment.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Post-Approval Commitment Study) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Targeted Enrollment (\$N\$):** 4 patients (originally planned for approximately 10 patients) **Number of Sites:** Multicenter (Taiwan) **Estimated Study Start:** 2022 **Estimated Primary Completion:** 31-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult male or female patients ≥ 18 years of age. 2. Participants with TSC associated with renal AML which is eligible for treatment with everolimus per local reimbursement criteria. 3. Signed informed consent must be obtained prior to participation in the study.

4.2 Exclusion Criteria 1. Patients with severe hepatic impairment (Child-Pugh class C). 2. Any severe and/or uncontrolled medical conditions. 3. Pregnant or breast-feeding females. 4. Patients with hypersensitivity to the active substance (everolimus), to other rapamycin derivatives, or to any of the excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Everolimus (Trade Names: Votubia, Zortress, Afinitor) - administered according to local prescribing criteria. **ATC Codes:** L01EG02, L04AA18 **Route of Administration:** Oral. **Duration of Treatment:** Up to 52 weeks of treatment phase. **EMA Reference URL:** [Afinitor EPAR Product Information](#)

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for medical conditions not restricted by the protocol. **Prohibited:** Concurrent treatment with other rapamycin derivatives or other investigational therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -30 to 0	Informed Consent, Medical History, AML Volume Imaging, Baseline Labs
Baseline	Day 1	Enrollment, First Dose, Laboratory Assessments
Interim	Up to Week 52	Continuous Safety Monitoring (AE/SAE), Laboratory Assessments (hematology, biochemistry), Efficacy Assessment (Imaging)
End of Study	Week 52 + 4 weeks	EOT (End of Treatment) Visit, 4-Week Safety Follow-Up Phase (for patients not continuing treatment beyond 52 weeks)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of participants experiencing adverse events (AEs), Serious adverse events (SAEs), and AEs of special interest (AESIs) over the study duration. **Measurement Method:** Clinical evaluation and central/local laboratory analysis.

7.2 Secondary & Exploratory Endpoints 1. **AML Response Rate:** Defined as a reduction in AML volume of at least 50% relative to screening (sum of target AML volumes). Additionally, no new AML ≥ 1 cm, kidney volume increases by no more than 20% from nadir, and absence of grade ≥ 2 AML-related bleeding. 2. **AML Progression Rate:** Evaluated as the percentage of participants experiencing progression of their angiomyolipomas. 3. **Laboratory Abnormalities:** Percentage of participants with newly occurring or worsening laboratory abnormalities.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection continuously throughout the 52-week treatment phase and the 4-week safety follow-up phase. **Serious Adverse Events (SAEs):** Reported promptly (typically within 24 hours) as per GCP standards. **Data Monitoring Committee (DMC):** Supervised internally by Novartis clinical safety and medical monitoring teams.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose) and Efficacy/Intent-to-Treat Set. **Sample Size Justification:** $N=4$ targeted enrollment; based on the pragmatic expectations of a post-approval commitment (PAC) study in a rare disease population (TSC-AML) within Taiwan. **Handling of Missing Data:** Missing components for volume assessments will be handled according to standardized tumor/lesion response criteria imputation rules.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring and data review by Novartis or its designated CRO. **Audit Trail:** Electronic Case Report Form (eCRF) utilized for locking data, alongside data clarification forms during clinical data management.

11. Study Identifiers & Governance

Primary ID: CRAD001M2402 **Registry Number:** NCT05252585 **Sponsor/Lead Organization:** Novartis **Study Title:** A Phase IV Study of Safety and Efficacy of Everolimus in Taiwanese Patients With Tuberous Sclerosis Complex Who Have Renal Angiomyolipoma (TSC-AML) **Official Regulatory Title:** Phase IV, Prospective Single Arm Study of Safety and Efficacy of Votubia (Everolimus) in Taiwanese Adults With Tuberous Sclerosis Complex Who Have Renal Angiomyolipoma

12. Clinical Framework (TSC-AML Specific)

Primary Condition / Disease Area: Renal Angiomyolipoma **Preferred Disease Name:** Angiomyolipoma of kidney (UMLS) **Diagnosis Threshold:** Confirmed diagnosis of TSC-AML fulfilling local reimbursement criteria **Medical Methodology:** Mammalian target of rapamycin (mTOR) inhibitor therapy **Optimization Target:** AML volume reduction $\geq 50\%$

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for TSC-AML medical therapy **Education Protocol (HCP):** PAC-specific protocol training and local reimbursement documentation guidance **Education Protocol (Patient):** Informed consent, AE reporting instructions, and visit adherence guidance **Quality Audit Mechanism:** Periodic laboratory assessment reviews and centralized imaging checks

14. Observation & Follow-Up Schedule

Total Duration: Up to 56 weeks (52 weeks of treatment + 4 weeks safety follow-up) **Onsite Visit Cadence:** Routine intervals defined by clinical management during the 52-week treatment period **Remote Monitoring Frequency:** Regular AE/SAE reporting tracking **Checklist Review Interval:** Laboratory assessment updates based on changes in abnormality grading

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				